

BotXRGame — Development Setup Log

Session Notes: Environment Setup, Unity MR Project, GitHub, ADB Connection

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1. Project Overview

BotXRGame is a mixed reality robot teleoperation game. A player wearing a Samsung Galaxy XR headset controls a physical robot in a real-world play area. Virtual game elements (obstacles, rewards) are overlaid on the real world using MR passthrough. The robot also navigates autonomously for parts of the game.

Component	Details
XR Headset	Samsung Galaxy XR
Robot Controller	RubikPi (Qualcomm) or equivalent
Game Engine	Unity 6000.4.2f1 — Mixed Reality template
Render Pipeline	URP (Universal Render Pipeline)
XR Framework	OpenXR + Android XR (com.unity.xr.androidxr-openxr 1.3.1)
XR Input	Hand Tracking only (Hand Interaction Profile)
ROS Version	ROS 2 Humble (Ubuntu 22.04)
Unity-ROS Bridge	ROS TCP Connector 0.7.0
Graphics API	Vulkan
Version Control	Git + GitHub (swapnil-0/BotXRGame)

2. GitHub Repository Setup

2.1 Repository Created on GitHub

Repo name: BotXRGame | Visibility: Private | License: MIT | .gitignore: Unity template

URL: <https://github.com/swapnil-0/BotXRGame>

2.2 Git Installation and Configuration

Installed Git for Windows from git-scm.com. Used Windows Terminal with Git Bash tab as primary terminal.

```
git config --global user.name "swapnil-0"
git config --global user.email "your@email.com"
```

2.3 Clone and Folder Structure

```
cd ~/Documents
git clone https://github.com/swapnil-0/BotXRGame.git
cd BotXRGame
mkdir unity
mkdir -p ros2_ws/src
```

```
touch unity/.gitkeep
touch ros2_ws/.gitkeep
```

Resulting folder structure:

```
BotXRGame/
■■■ unity/      <- Unity MR project
■■■ ros2_ws/src/ <- ROS 2 workspace
■■■ docs/       <- Shared documentation
```

2.4 First Commit and Push

```
git add .
git commit -m "Initial folder structure: unity and ros2_ws"
git push origin main
```

3. Unity Project Setup

3.1 Project Creation

Created Unity project using Mixed Reality (MR) template in Unity Hub 6000.4.2f1. Location set to:
C:\Users\swapn\Documents\BotXRGame\unity\

3.2 Editor Settings (Before First Commit)

Edit -> Project Settings -> Editor:

Version Control Mode: Visible Meta Files

Asset Serialization Mode: Force Text

Note: Mixed Reality template already had these set correctly by default.

3.3 Packages Installed

Package	Version	How Added
ROS TCP Connector	0.7.0-preview	Git URL
Unity OpenXR Android XR	1.3.1	Git URL
OpenXR Plugin	1.17.1	Unity Registry
XR Interaction Toolkit	Latest	Unity Registry
XR Device Simulator	Included	Samples tab
Starter Assets	Included	Samples tab

ROS TCP Connector Git URL:

```
https://github.com/Unity-Technologies/ROS-TCP-Connector.git?path=/com.unity.robotics.ros-tcp-connector
```

3.4 XR Configuration

Edit -> Project Settings -> XR Plug-in Management -> Android tab:

- Enabled: OpenXR
- Added Interaction Profile: Hand Interaction Profile
- Enabled Feature: Hand Tracking Subsystem

3.5 URP Material Conversion

Imported Sci-Fi Space Fighter PBBRP asset (free, URP compatible) from Unity Asset Store by DUNGEON MASON. Materials appeared pink (Built-in pipeline). Fixed using:

Window -> Rendering -> Render Pipeline Converter -> Built-in to URP -> Material Upgrade -> Initialize Converters -> Convert Assets

3.6 Android XR Build Profile

File -> Build Profiles -> Add Build Profile -> XR -> Android XR -> Switch Profile

Graphics API confirmed: Vulkan (first in list)

Bundle Identifier set in Build Profile Player Settings: com.swapnil.BotXRGame

4. Git Configuration for Unity

4.1 .gitignore Fix

The root .gitignore was not covering the Unity project subfolder. Copied it to the Unity project folder:

```
cp ~/Documents/BotXRGame/.gitignore \  
~/Documents/BotXRGame/unity/BotXRGame/.gitignore
```

Key folders excluded from Git: Library/, Temp/, Logs/, UserSettings/

4.2 Unity Project Commit

```
cd ~/Documents/BotXRGame  
git add .  
git commit -m "Unity MR project setup - packages and XR configuration"  
git push origin main
```

Push size: ~235MB (first Unity project commit)

5. Documentation Added to Repo

Created docs/ folder structure and two key documents:

```
mkdir -p docs/architecture docs/ros-interface docs/presentations
```

File	Purpose
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docs/architecture/system-overview.md	Full system architecture, game flow, team roles, hardware/software stack, current
docs/ros-interface/ros-unity-interface.md	All ROS 2 topics, message types, directions, setup instructions for ROS Dev

```
git add .
git commit -m "Add system architecture and ROS-Unity interface documentation"
git push origin main
```

6. ADB Connection to Galaxy XR Headset

6.1 USB Connection — Failed

USB-C connection to PC failed with 'device not recognized'. Installed Samsung USB Driver V1.9.5.0 but USB still not recognized. Root cause: cable may be charge-only or driver compatibility issue.

6.2 Wireless ADB — Success

Both PC (ethernet) and Galaxy XR (WiFi) on same router subnet 192.168.1.x confirmed by:

```
ipconfig | grep "IPv4"
# PC IP:
# Headset IP:
```

Enabled Wireless Debugging on headset: Settings -> Developer Options -> Wireless Debugging -> ON

Added Windows Firewall rule (run Git Bash as Administrator):

```
netsh advfirewall firewall add rule name="ADB" dir=in action=allow protocol=TCP localport=5037
netsh advfirewall firewall add rule name="ADB_out" dir=out action=allow protocol=TCP localport=5037
```

ADB path used (Unity's bundled ADB):

```
cd "/c/Program Files/Unity/Hub/Editor/6000.4.2f1/Editor/Data/PlaybackEngines/AndroidPlayer/SDK/platform-tools"
```

Paired device using Wireless Debugging -> Pair with pairing code:

```
./adb pair :
# Enter pairing code:
# Successfully paired to : [guid=]
```

Connected to headset:

```
./adb connect :
# connected to :

./adb devices
# List of devices attached
# : device
```

Device appeared in Unity Build Profiles as: Samsung SM I610 (:)

7. Current Status

- ✓ GitHub repo created (swapnil-0/BotXRGame)
- ✓ Git installed and configured on PC
- ✓ Folder structure set up (unity/, ros2_ws/, docs/)
- ✓ Unity 6000.4.2f1 MR project created
- ✓ ROS TCP Connector installed (0.7.0)
- ✓ Unity OpenXR Android XR installed (1.3.1)
- ✓ OpenXR Plugin installed (1.17.1)
- ✓ XR Interaction Toolkit + Samples installed
- ✓ Hand Interaction Profile configured
- ✓ Android XR Build Profile active
- ✓ Vulkan confirmed as graphics API
- ✓ gitignore correctly excluding Library/Temp/Logs
- ✓ System architecture doc added to GitHub
- ✓ ROS-Unity interface spec doc added to GitHub
- ✓ Sci-Fi Space Fighter asset imported (URP converted)
- ✓ Galaxy XR headset connected via wireless ADB

Next Steps

- Fix Bundle ID in Build Profile Player Settings and complete first build to headset
- Build and deploy current scene to Galaxy XR for first on-device test
- Add ROSConnection GameObject to Unity scene
- Implement virtual joystick with hand tracking pinch input
- Add bot movement script driven by joystick values
- Add HUD panel showing live cmd_vel ROS message values
- Install ROS 2 Humble on RubikPi
- Configure and test ros_tcp_endpoint
- Test full Unity-to-ROS communication over WiFi